

EXPERIMENT 16: FEED FORWARD NEURAL NETWORK

Aim

To write a Python program to implement a **Feed Forward Neural Network** for classification.

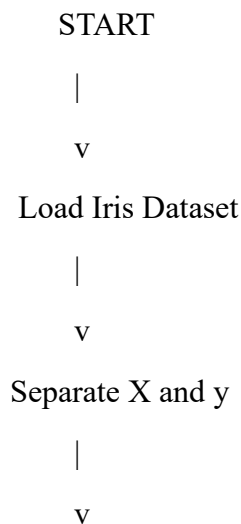
Objective

- To understand the architecture of a neural network.
- To implement input, hidden, and output layers.
- To train a neural network using sample data.
- To predict the class of unseen data.
- To evaluate the performance using accuracy.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate input features and target classes.
4. Split the dataset into training and testing data.
5. Create a Feed Forward Neural Network using MLPClassifier.
6. Define the hidden layer.
7. Train the network using training data.
8. Predict the test data.
9. Calculate the accuracy.
10. Display the prediction and accuracy.

Flowchart



Split Train/Test

|

v

Create Neural Network

|

v

Input Layer

|

v

Hidden Layer

|

v

Output Layer

|

v

Train the Network

|

v

Predict Test Data

|

v

Calculate Accuracy

|

v

Display Result

|

v

END

Neural Network Architecture

INPUT LAYER

4 Features

|

v

HIDDEN LAYER

10 Neurons

|

v

OUTPUT LAYER

3 Classes

|

v

Prediction

Python Program

Writing

Feed Forward Neural Network

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
```

```
test_size=0.2,
random_state=42
)

# Create Feed Forward Neural Network
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    max_iter=1000,
    random_state=42
)

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Actual values:")
print(y_test)

print("Predicted values:")
print(y_pred)

print("Accuracy:", accuracy)

# Sample prediction
sample = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(sample)
```

```
print("Sample Prediction:",  
      iris.target_names[prediction][0])
```

Sample Output

Actual values:

```
[1 0 2 1 1 0 1 2 1 1 ...]
```

Predicted values:

```
[1 0 2 1 1 0 1 2 1 1 ...]
```

Accuracy: 0.9666666666666667

Sample Prediction: setosa

Result

Thus, the **Feed Forward Neural Network** was successfully implemented using the **Iris dataset** and the **classification accuracy** was obtained.